Christopher Nolan wanted the wormhole to have a mild gravitational pull. Strong enough to hold the *Endurance* in orbit around itself, but weak enough that a modest rocket blast would slow the *Endurance*, letting it drop gently into the wormhole. This meant a gravitational pull much less than the Earth's.

Einstein's law of time warps tells us that the slowing of time inside the wormhole is proportional to the strength of the wormhole's gravitational pull. With that pull weaker than the Earth's pull, the slowing of time must be smaller than on Earth, which is only a part in a billion (that is, one second of slowing during a billion seconds of time, thirty years). Such slowing is so tiny that Oliver and I paid no attention to it at all when designing the wormhole.

## "Handles" for Adjusting the Wormhole's Shape

The ultimate decision about the wormhole's shape was in the hands of Christopher Nolan (the director) and Paul Franklin (the visual-effects supervisor). My task was to give Oliver and his colleagues at Double Negative "handles" (or in technical language, "parameters") that they could use to adjust the shape. They then simulated the wormhole's appearance for various adjustments of the handles and showed the simulations to Chris and Paul, who chose the one that was most compelling.

I gave the wormhole's shape three handles—three ways to adjust the shape (Figure 15.1).